Measuring the Value of Enterprise Records in Risk-Controlled Invoice Automation

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***Abstract*—Many procure-to-pay systems post an invoice without human review once its extraction confidence passes a fixed threshold, but the error rate this admits is seldom measured. It is also unclear whether checks against the organization's own records would let more invoices skip review at the same error rate. We study both questions on a seeded synthetic procure-to-pay world of 100 vendors, 1,649 purchase orders and 3,000 invoices, 29.8 percent of which carry one injected business error, read with the extraction code of a deployed platform. We compare the platform's confidence rule, a rule that accepts when every record check passes, and risk-controlled gates that set their threshold with Learn-then-Test from document evidence alone or from document and record evidence, over 20 vendor-disjoint splits. The deployed rule accepts 41.2 percent of invoices, and 7.2 percent of those are wrong. Record evidence improves ranking, raising the area under the ROC curve from 0.735 to 0.945, but no gate specified in advance certifies more than 2.6 percent of invoices at error budgets of 1, 2 or 5 percent, so our hypothesis of a five-point coverage gain is not supported. Most top-ranked errors are resubmissions under a new invoice number, which the production duplicate rule misses. A duplicate check that ignores the number, defined after this diagnosis, certifies 21.0 and 37.0 percent of invoices at the 2 and 5 percent budgets on three new seeds, but exceeds the error budget in up to 20 percent of splits. All data are synthetic.**

***Index Terms*—selective prediction, risk control, Learn-then-Test, invoice processing, procure-to-pay, straight-through processing**

I. Introduction

An automated procure-to-pay (P2P) pipeline reads each incoming invoice, links it to a vendor and a purchase order (PO), and then either posts it or sends it to a person. Posting without review is a *selective prediction* decision: the system trades coverage, the share of documents handled automatically, against the error rate among the documents it accepts [1], [2]. Work on document extraction usually reports field accuracy on benchmarks such as SROIE [3] or LayoutLMv3 [4]. A posting gate is judged differently: what counts is how many accepted invoices are wrong, and an invoice can be wrong even when every field was read correctly. It may be a resubmission, carry changed bank details, or bill more than was ordered.

Learn-then-Test (LTT) [5] chooses a threshold so that the error rate among accepted items stays below a target with high probability. Recent work brings calibrated confidence and conformal guarantees to straight-through processing [6], [7], but scores each field on what the document itself contains. Commercial P2P tools do match invoices against vendor masters and POs [8]–[10], with hand-set tolerances and no published error rates.

This paper starts from the review gate of a procurement platform we built [11]. It accepts an invoice when a weighted extraction confidence is at least 0.8 and no review flag is raised. We ask four research questions (RQs):

- **RQ1.** What error rate does this fixed rule admit on an invoice stream that contains business errors?
- **RQ2.** Does evidence from the organization's records let a risk-controlled gate accept more invoices at the same error budget? Our hypothesis (H1), fixed before any experiment, was a gain of at least 5 percentage points over a document-only gate at a 2% budget.
- **RQ3.** How much more calibration data does per-field risk control need than whole-document control?
- **RQ4.** How much of the ranking signal survives when the records are out of date?

The contributions are empirical. We do not propose a new risk-control method or a new extractor. (1) We measure the error rate of the deployed rule, and of a rule that trusts every record check, on a P2P benchmark with known ground truth. (2) We compare risk-controlled gates with and without record evidence over vendor-disjoint splits, with an ablation per check and a stale-records test. (3) We explain why no gate specified in advance certifies a useful share, and report a post hoc fix, tested on new seeds, together with what it needs before it can be trusted.

II. Related Work

A. Selective Prediction and Risk Control

Selective classification lets a model abstain to lower its risk. El-Yaniv and Wiener describe the resulting risk–coverage trade-off for noise-free models [1], and Geifman and El-Yaniv build a selective classifier on top of a trained network so that a chosen risk level holds with high probability [2]. LTT treats the choice of a low-dimensional parameter, such as a threshold, as a multiple-testing problem and gives finite-sample, high-probability guarantees for any underlying model [5]. Conformal risk control instead bounds the expected value

of a loss that is monotone in the parameter [12]. Selective risk is not monotone in the threshold, so we use LTT. The guarantee holds whatever the quality of the score; the coverage obtained depends on how well the score separates correct from incorrect items, which RQ2 varies.

Learning to defer models the human who receives rejected items [13], [14]; we treat review as a fixed, error-free fallback. The platform's field confidences are hand-set, and even learned confidences are often miscalibrated [15], so we learn a probability model before thresholding. LTT assumes that calibration and deployment data are exchangeable; weighted conformal prediction relaxes this under covariate shift [16], but we do not apply it (Section VII).

B. Document Extraction and Its Benchmarks

Layout-aware models read business documents well [4], and optical character recognition (OCR) engines such as PaddleOCR and Docling are widely used [17], [18]. The benchmarks measure reading. SROIE covers receipts [3]; DocILE [19] and FATURA [20] cover invoices. None of them ships the purchase orders, vendor master or payment history an invoice belongs to, so none can say whether posting it was right. Krieger *et al.* show that extractors degrade on rarely seen suppliers [21], which bears on unseen test vendors.

C. Confidence for Straight-Through Processing

HYCEDIS scores extracted key–value pairs with a conformal predictor and an anomaly detector [22]; Rombach and Mehdiyev build entity-level conformal sets for receipts [7]; ConfBench measures how well vision–language model confidence tracks extraction accuracy [23]. The closest work is that of Jin *et al.* [6], who fuse perception, layout and validation features with gradient boosting and set per-field LTT thresholds on DocILE, FATURA and VRDU. Their validators are document-internal and add little signal on invoices, and their decisions are per field.

D. Procurement Practice and the Gap

Most AI deployments in procurement sit in the P2P cycle [24], [25], and three-way matching is standard in commercial tools [8]–[10] (Table I). Within the literature we searched, no study (i) measures the error admitted by a confidence-threshold gate on invoices with business errors, or (ii) tests whether enterprise records change what a risk-controlled gate can certify for the whole document. Practice already uses these checks as hard rules; what has not been measured is their value inside a gate with an error guarantee.

TABLE I
CLOSEST WORK COMPARED WITH THIS STUDY

Work	Guarantee	Records	Unit	Records' value
Jin *et al.* [6]	LTT	no	field	n/a
Rombach, Mehdiyev [7]	conformal	no	entity	n/a
ConfBench [23]	none	no	entity	n/a
Rossum [8], [9]	none	rules	field	unmeasured
SAP Ariba [10]	none	rules	invoice	unmeasured
This study	LTT	yes	invoice	ablation

From published text and documentation, Oct. 2026.

TABLE II
FEWEST ERROR-FREE ACCEPTED CALIBRATION INVOICES LTT NEEDS ($\boldsymbol{\delta = 0.1}$)

Gate	$\alpha = 1\%$	$\alpha = 2\%$	$\alpha = 5\%$	$\alpha = 10\%$
Whole document	230	114	45	22
Per component ($\alpha/5$)	1,151	575	230	114

From (5). A calibration set holds the invoices of 25 vendors, about 750 invoices on average.

III. PROBLEM FORMULATION

Each invoice i has document evidence D_i (extracted fields, field confidences, text quality, classification confidence, internal consistency checks), record evidence E_i (results of checks against the vendor master, open POs and earlier invoices), and a label $Y_i \in \{0, 1\}$. We set $Y_i = 1$ if and only if the vendor, total, currency and invoice number were read correctly *and* no business error was injected. These four fields decide who is paid, how much and against which document; a misread date or PO number does not by itself make the payment wrong. A gate is a score $s(D, E) \in [0, 1]$ and a threshold λ; the invoice is accepted without review if and only if $s \geq \lambda$. Coverage and selective risk are

$$\Phi(\lambda) = \Pr[s(D, E) \geq \lambda]\,, \tag{1}$$

$$\mathcal{R}(\lambda) = \Pr[Y = 0 \,|\, s(D, E) \geq \lambda]\,. \tag{2}$$

For an error budget α and a confidence parameter δ, we want the threshold with the largest coverage such that

$$\Pr\Big[\mathcal{R}(\hat{\lambda}) \leq \alpha\Big] \;\geq\; 1 - \delta, \tag{3}$$

where the probability is over the calibration sample. LTT tests the null hypotheses $H_k : \mathcal{R}(\lambda_k) > \alpha$ on a grid $\lambda_1 > \lambda_2 > \cdots > \lambda_K$ ordered from strict to loose. If n_k calibration invoices are accepted at λ_k and e_k of them are wrong, the p-value is the binomial tail

$$p_k = \Pr[\mathrm{Bin}(n_k, \alpha) \leq e_k]\,. \tag{4}$$

Fixed-sequence testing rejects $H_1, H_2, \ldots$ while $p_k \leq \delta$ and stops at the first failure. This controls the family-wise error at δ, so the loosest rejected threshold satisfies (3) [5]. If H_1 is not rejected, nothing is accepted. Even with no errors ($e_k = 0$), rejection needs

$$n_k \;\geq\; \log\delta / \log(1 - \alpha) \tag{5}$$

accepted calibration invoices. Table II lists this floor for the budgets we study, for a whole-document gate and for a per-component gate that runs at $\alpha/5$. It matters in Section VI.

IV. METHODOLOGY

A. Deployed Rule and Extraction

In the study, text PDFs are read with a text-layer parser and scans with PaddleOCR 2.9.1 using the platform's settings and line parsing [17]; a TF-IDF and logistic-regression classifier

TABLE III
SYNTHETIC BENCHMARK (GENERATOR 1.0, SEED 7)

Item	Value
Vendors / purchase orders / invoices	100 / 1,649 / 3,000
Invoices with an injected error	895 (29.8%)
duplicate (renumbered resubmission)	147
unit price above the PO	145
changed bank details	117
quantity above the PO's remainder	109
total not equal to subtotal plus tax	109
look-alike vendor name	95
another vendor's PO cited	88
wrong or invalid GSTIN	85
Invoices with all four label fields read correctly	84.1%
Correct invoices ($Y = 1$)	59.8%
Share / fields read correctly / correct invoices[a]	
text PDF	55.1 / 100 / 72.3
clean scan	25.5 / 91.2 / 63.0
mild scan	11.5 / 50.0 / 34.1
heavy scan	7.9 / 0.0 / 0.0

[a]All three values in percent.

assigns the document type and regular expressions extract the fields. (The live pipeline also uses Docling [18] and LayoutLMv3; neither is part of this study.) Each field f has a confidence $c_f \in [0,1]$, discounted by the quality $q \in [0,1]$ of the text it was read from. The document score is a weighted mean:

$$s_f = c_f\, q, \quad S = \frac{\sum_{f \in F} w_f\, s_f}{\sum_{f \in F} w_f}, \quad \text{review if } S < \theta = 0.8, \tag{6}$$

with w_f = 0.30 (total), 0.20 (vendor name), 0.15 (classification), 0.15 (document number), 0.10 (date) and 0.10 (line items, left out when none are found) [11]. The deployed rule also forces review when the vendor name is missing or only resembles a known vendor, when the document is not an invoice, when the PO match fails, when a duplicate is found, or when bank details differ from the master. Confidence and these flags together form baseline Conf.

B. Synthetic Procure-to-Pay World

A seeded generator builds a vendor master of 100 vendors, each with a Goods and Services Tax Identification Number (GSTIN), bank details and a currency (21 of the 100 vendors invoice in US dollars, the rest in Indian rupees, INR). Invoice volume follows a Zipf-like weight $1/(1+i)^{0.8}$ over vendors ranked $i = 0, \ldots, 99$, so a few vendors send most invoices; the largest sends 350 of the 3,000. The generator then issues 1,649 POs and 3,000 invoices against them. Each invoice carries at most one injected business error of eight types, listed with their counts in Table III. A resubmitted duplicate copies an earlier invoice of the same vendor, moves its date by one to three days, and always takes the vendor's next invoice number. Invoices are rendered as text PDFs or as clean, mild or heavily degraded scans; ground truth is known by construction.

So 40.2% of invoices are incorrect: 29.8 points from injected errors and 10.4 points from misread fields alone. Heavy scans are never read correctly, so every gate rejects them; reading errors are hard to spot on mild scans.

C. Record Checks

The checks reuse the platform's rules. Vendor matching compares the normalized name with the master using the token-sort ratio of RapidFuzz 3.9.6; a score of 88 or more is a match and 70 to 87 a look-alike. The three-way match compares the invoice's lines with the vendor's open POs and holds the invoice when a unit price or the remaining balance is exceeded by more than 1%. Bank details, GSTIN and currency are compared with the master. Two invoices i, j of the same vendor are duplicates when

$$|a_i - a_j| \le 0.02\, a_j \;\wedge\; |d_i - d_j| \le 7 \text{ days} \;\wedge\; \mathrm{sim}(n_i, n_j) \ge 0.85, \tag{7}$$

where a is the amount, d the date, n the invoice number and sim a string-similarity ratio. All three limits are configurable business rules [11]. Invoice numbers have the form INV-<vendor>-<counter>, such as INV-025-0003; between an injected duplicate and its original, sim has median 0.83 and is below 0.85 for 50.3% (74) of the 147 duplicates. The exploratory check of Section VI-D drops the last condition, which equals setting that rule to zero. The checks add about 1–2 ms per invoice (measured in a Docker VM with 15 vCPUs and 7.7 GB on an Apple M5 Pro laptop).

D. Gate Systems

We compare six systems:

- Conf: the deployed rule of Section IV-A.
- Conf+LTT: the deployed score S of (6) with the same review flags, but with its threshold set by LTT instead of fixed at 0.8.
- Rules: accept when every record check passes (vendor matched and not a look-alike, PO matched, no duplicate, and bank details, GSTIN and currency agree with the master where present). It ignores extraction confidence.
- G-doc: a probability model of $\Pr[Y = 1]$ on 19 document features, thresholded with LTT.
- G-rec: the same on the document features plus 16 record features (vendor, PO, duplicate, bank, GSTIN, currency, vendor history).
- G-comp: one LTT gate per payment-critical component (vendor, total, currency, number, business error), each at $\alpha/5$; an invoice is accepted when all five accept. This mirrors per-field designs (RQ3) but does not certify document-level risk: each gate bounds its component's error among the invoices *it* accepts, not among those all five accept.

The probability model is either a standardized logistic regression or histogram gradient boosting (scikit-learn 1.9.1 [26]; 300 iterations, learning rate 0.05, 15 leaves, L2 penalty 1.0). G-comp uses logistic regression. The full feature lists are in the code release.

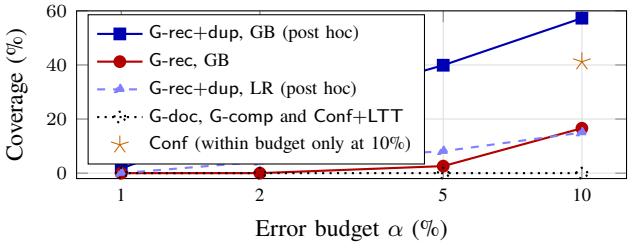


Fig. 1. Coverage each system achieves while meeting the budget (means over 20 splits; GB: gradient boosting, LR: logistic regression). Rules meets none of the budgets. The learned gates specified in advance accept almost nothing below 10%.

V. Experimental Setup

Each of 20 random splits assigns vendors 50/25/25 to training, calibration and test, so test vendors are never seen before. The model is fitted on training vendors. Candidate thresholds come from quantiles of the in-sample training scores at 40 coverage levels, starting at 1.5 times the floor (5) (as a share of the calibration set); the grid uses the calibration set only through its size, never through its scores or labels. LTT then picks a threshold on the calibration vendors with $\delta = 0.1$. The plan fixed error budgets of 1%, 2% and 5%; we added 10% after nothing passed at 5%.

For each system we report means over splits of coverage (1) and risk (2) on the test vendors, and the *violation rate*, the share of splits with test risk above α, which should be at most δ. Mean risk is the mean over all 20 splits, a split that accepts nothing counting as zero risk, so it understates the risk of a gate that accepts in few splits; we then also give the risk over the splits that accept. A gate that accepts nothing in any split has its risk shown as a dash. For the learned gates we also report the area under the ROC curve (AUROC) and under the risk–coverage curve (AURC). We give the value in INR of wrongly accepted invoices, converting US-dollar invoices at 83 INR per dollar.

Coverage differences between systems get a paired percentile bootstrap 95% confidence interval (CI) over splits, which describes variation across splits, not the sampling uncertainty of the generator (the splits reuse the same 100 vendors). Violation rates get exact Clopper–Pearson 95% intervals, since a test set of about 750 invoices makes them noisy even when the guarantee holds.

For RQ4 we make a share of the records stale (0 to 40%): those vendors' bank details change and their GSTIN goes missing, and the same share of POs is never entered. The model is trained on equally stale records. One script writes all figures to a results file with the generator version, seed and commit. Code and generator: https://github.com/AnoojShete/ProcurementAutomation.

VI. Results and Discussion

Table IV gives the main results, and Fig. 1 plots coverage across all four budgets.

TABLE IV
Review Gates on 3,000 Synthetic Invoices, Means over 20 Vendor-Disjoint Splits

System	$\alpha = 2\%$		$\alpha = 5\%$		AUROC	AURC
	Cov.	*Risk*	*Cov.*	*Risk*		
Conf[a]	41.2	7.2	41.2	7.2	–	–
Conf+LTT	0.0	–	0.0	–	0.851	0.171
Rules[a]	58.8	13.2	58.8	13.2	–	–
G-doc[b]	0.0	–	0.0	–	0.735	0.268
G-rec[b]	0.0	–	2.6	0.6	0.945	0.127
G-comp	0.0	–	0.0	–	–	–
G-rec+dup[b,c]	20.7	0.7	39.9	2.9	0.971	0.107

Cov.: percent of test invoices accepted without review. Risk: percent of accepted invoices that are wrong. At $\alpha = 1\%$ every gate accepts nothing. [a]Fixed rules: they violate the 2% budget in all splits and the 5% budget in 90% (Conf) and 100% (Rules) of splits. [b]Gradient boosting. With logistic regression the AUROCs are 0.745, 0.940 and 0.958, G-rec accepts nothing, and G-rec+dup covers 4.2% and 8.1% (risk at 5%: 0.8%). [c]Defined post hoc (Section VI-D); violation rate 15% at $\alpha = 2\%$ and 15% at $\alpha = 5\%$.

A. Error Rate of the Deployed Rule

Conf accepts 41.2% of invoices (standard deviation 1.2 points across splits), and 7.2% of those are wrong, so about 3.0% of all invoices would be paid in error. The wrongly accepted invoices are worth INR 15.8 million (158.3 lakh) per test set, which holds about 750 invoices on average. Trusting the record checks alone (Rules) covers more, 58.8%, at almost twice the risk (13.2%, INR 39.3 million). Conf already raises most of these checks as review flags, so its errors pass the flags as configured. Neither rule meets a 2% or 5% budget (Table IV). Conf does meet a 10% budget in every split (its highest split risk is 9.0%). Over all 3,000 invoices, Conf wrongly accepts 93 of the 1,232 it accepts: 46% carry a wrong GSTIN and 41% are renumbered duplicates. Rules wrongly accepts 248 of 1,766: 34% have an arithmetic error, 25% a misread field, and 22% are duplicates.

Tuning the threshold of the deployed score with LTT (Conf+LTT) gives no coverage at either budget: it accepts nothing in all 20 splits (AUROC 0.851, AURC 0.171). Tuning θ alone therefore does not fix the rule, because the errors it admits, wrong GSTINs and renumbered duplicates, are not ones that extraction confidence can see.

B. Value of Record Evidence

Record evidence raises AUROC from 0.735 to 0.945 with gradient boosting and from 0.745 to 0.940 with logistic regression, and halves AURC (0.268 to 0.127). The PO match carries most of this. Removing it lowers the logistic AUROC from 0.940 to 0.833; removing bank details, the duplicate check or the GSTIN check gives 0.898, 0.921 and 0.923, while vendor matching, currency and history change it by at most 0.002. The three-way match runs only for a matched vendor, so the PO features carry most of what vendor matching adds.

Better ranking did not become coverage at the planned budgets. G-rec certifies nothing at 1% and 2%, and 2.6% at 5% (mean risk 0.6% over all splits, 4.3% over the 3 splits that accept; one violation), accepting invoices in only those 3 of the 20 splits; G-doc certifies nothing. The paired gain of

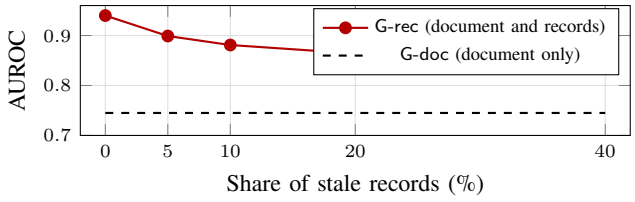


Fig. 2. Ranking signal as bank details, GSTINs and POs go out of date (logistic regression, means over 20 splits). With 40% of the records stale, G-rec still ranks better than G-doc.

G-rec over G-doc is $+0.0$ points at $\alpha = 2\%$ and $+2.6$ points (95% CI $[0.0, 5.9]$) at 5%, so H1 is not supported. The gain appears only at the unplanned 10% budget: G-rec then covers 16.6% at 2.4% risk and a 5% violation rate (1 of 20 splits; CI 0.1–24.9%), a gain of $+16.6$ points (CI $[5.9, 28.6]$) over G-doc (Fig. 1).

C. Why the Gates Certify Little

At a 2% budget, LTT needs at least 114 accepted calibration invoices with almost no errors (Table II). Among the top 16.2% of calibration invoices ranked by G-rec (mean over splits), the smallest slice that reaches this count, 3.8% are wrong, so no threshold passes and LTT accepts nothing. Of the 86 wrong invoices in these slices (summed over splits), 67 (78%) are renumbered resubmissions. The duplicate rule (7) needs the two invoice numbers to be at least 85% similar, which half of the new numbers are not (Section IV), and the PO still has quantity left, so the three-way match passes too. The rest are overbilled quantities (7), arithmetic errors (5), misread fields (4) and overbilled prices (3). The limit lies mainly in one check, not in the model.

D. Exploratory Number-Independent Duplicate Check

Adding a feature for the duplicate check without its number condition gives G-rec+dup. With gradient boosting it certifies 20.7% of invoices at $\alpha = 2\%$ (risk 0.7%) and 39.9% at 5% (risk 2.9%), with AUROC 0.971. It accepts in 11 of 20 splits at 2% (risk 1.2% among them) and in 18 at 5% (3.2%). Its violation rate is 15% at both budgets (3 of 20 splits; CI 3.2–37.9%) and 20% at the unplanned 10% budget (4 of 20; CI 5.7–43.7%). All three point estimates are above $\delta = 0.1$, although with 20 splits the intervals do not exclude it. The error guarantee is therefore not met reliably. On correct invoices the check without the number condition fires on 4.6% (82 of 1,794); the generator creates some legitimate repeat invoices with the same amount within seven days (26 of 2,105 clean invoices), so this false alarm rate may be optimistic.

The gate was defined after we saw the diagnosis, on the same seed. We therefore froze its definition (commit 1306b21) and ran it unchanged on 3 new generator seeds, where it covers $21.0 \pm 10.3\%$ at $\alpha = 2\%$ and $37.0 \pm 7.0\%$ at 5% (mean $\pm$ SD over the three seeds). Its violation rate on those seeds is 20, 10 and 0% at 2% and 5, 10 and 10% at 5%. In a generator variant where 50% of duplicates keep the original number or get a one-character edit, its coverage on seed 7 is 20.3% and 42.0% (violation rate 15% at both budgets).

TABLE V
LARGEST GROUPS OF WRONGLY ACCEPTED INVOICES, G-REC+DUP WITH GRADIENT BOOSTING AT $\alpha = 10\%$, SUMMED OVER 20 SPLITS

Injected error	Rendering	Misread label field	Count
total $\neq$ subtotal plus tax	clean scan	none	112
none	mild scan	total	76
unit price above PO	clean scan	none	74
none	clean scan	number	63
quantity above PO	clean scan	none	41
duplicate	clean scan	none	38

Table V lists the largest groups of invoices that G-rec+dup still accepts wrongly at $\alpha = 10\%$, the one budget with enough accepted invoices to show a pattern. Arithmetic errors on clean scans lead. They are not missed by the extractor: its consistency check reports a mismatch on all 127 of them, but neither the deployed rule nor the learned gate treats that flag as a reason to stop. Next come correct invoices whose total or number was misread on a scan, and overbilled prices on clean scans. Renumbered duplicates still appear, mostly on clean scans. In 38 such cases the amount matched the original in all, but the date was not read at all in 26 and the numbers were less than 85% similar in 17, so the rule did not pair those invoices.

E. Per-Component Gates and Stale Records

G-comp accepts nothing at any budget. Table II explains why: at $\alpha/5$, each component gate needs at least 575 error-free accepted calibration invoices at a 2% budget and 230 at 5%, against about 750 calibration invoices in total, of which only 59.8% are correct. Per-component control therefore needs five times the calibration data of document-level control at the same budget (RQ3), and still would not certify document-level risk. A valid per-component design would score each invoice by its lowest component probability and run one LTT on document-level risk, keeping the whole-document floor.

With stale records (RQ4), the AUROC of G-rec falls steadily, from 0.940 to 0.899, 0.881, 0.864 and 0.838 as 5, 10, 20 and 40% of the records go stale (logistic regression; Fig. 2). It stays above the document-only 0.745 at every level. The logistic gate's coverage is near zero at all levels, so we repeat the test with gradient boosting at $\alpha = 10\%$, where G-rec covers 16.6% on current records: its coverage is 20.1, 8.1, 9.8 and 5.1% at 5, 10, 20 and 40% stale, and its violation rate 10, 0, 5 and 5% (5% on current records). Coverage falls by about two thirds at 40% stale records, but not monotonically; the means are noisy with 20 splits.

VII. LIMITATIONS

The invoices, records and error rates come from one generator and no real or public invoice data were used, so absolute coverage and risk will not transfer to another organization; the generator also decides how detectable each error is. Every injected duplicate carries a new number, so the gain from the

number-independent check partly reflects that design choice; the variant above suggests it is not the whole gain. The share of invoices with a business error, 29.8%, is far above what most organizations see, so coverage at a fixed budget would be higher at a lower prevalence. The planned run on public invoices with derived records has not been done. G-rec+dup was chosen after inspecting results.

Invoices of one vendor are not independent, but the binomial test (4) treats them as if they were; with one vendor sending 350 of 3,000 invoices, the effective calibration size is smaller than the count. This may explain the violation rates above δ of G-rec+dup: 15% at 2% and 5%, and 20% at the unplanned 10% budget. No correction for this, such as cluster-aware bounds or weighted calibration [16], was applied.

Each calibration set holds only 25 vendors, and we use a single δ. Reviewers are treated as free and always right [14]. Results depend on one extraction stack; the Docling and LayoutLMv3 paths were not exercised. Seed 7 was re-run on 6 Nov 2026 after the OCR image was rebuilt; the earlier run (19 Oct) gave somewhat different numbers, so absolute values depend on library versions.

VIII. Conclusion

On a synthetic P2P benchmark, a hand-set confidence rule accepted 41.2% of invoices with 7.2% of them wrong, above all three budgets we planned to certify. Record evidence improved the ranking of invoices substantially, mostly through the PO match, but no risk-controlled gate specified in advance certified more than 2.6% of invoices at budgets of 5% or less, and the hypothesized gain was not found. The main bottleneck was one rule: resubmitted invoices with a new number pass the duplicate check. A check that ignores the number certified 21.0% and 37.0% at the 2% and 5% budgets on three new seeds, but it was found by looking at the results and its error rate exceeded the budget in up to 20% of splits.

Next we will use public invoices (DocILE or FATURA) with derived records, cluster-aware bounds, and a model of reviewer cost.

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